Homework: Gardener Problem

	[,1]	[,2]	[,3]
[1,]	0.3	0.6	0.1
[2,]	0.1	0.6	0.3
[3,]	0.05	0.4	0.55

Consider the gardener problem with fertilizer. Suppose that the cost of the fertilizer is $50 per bag and the garden needs two bags if the soil is good. The amount of fertilizer is increased by 25% if the soil is fair and 60% if the soil is poor.

The gardener estimates that annual yield to be worth $250 if no fertilizer is used and $420 if fertilizer is applied. Is it worthwhile to use the fertilizer?

- Write balance equations to determine the steady state probabilities π_1, π_2, π_3
- Determine steady state probability of good soil
- Suppose fertilizer is applied.
  - Determine expected total cost.
  - Should fertilizer be applied?
- (Optional*) Determine the mean first return time to good soil.

State k	1: good	2: fair	3: poor
Cost of fertilizer $c(k)$	100	125	160

Homework: Bank inactivity collection fee

- Consider the steady state.
- Write down the steady-state balance equations.
- Let the probability that account is inactive next month be $r = 0.30$. Solve for the steady-state distribution.
- Recall that the fee is charged each month while the account remains in state 5.
  Let F be the monthly fee. Then, the long-run expected fee revenue per account per month is

  $$\mathbb{E}[fee] = F\,\pi_5 \qquad \text{(THB/account/month)}$$

  Compute the expected fee for F= 75 THB.
- (Optional) Derive a closed-form expression for $\pi_5(r)$.

State	Description
0	Active this month
1	1 month inactive
2	2 months inactive
3	3 months inactive
4	4 months inactive
5	More than 4 months inactive (5 months or more inactive)

$$P = \begin{bmatrix} 1-r & r & 0 & 0 & 0 & 0 \\ 1-r & 0 & r & 0 & 0 & 0 \\ 1-r & 0 & 0 & r & 0 & 0 \\ 1-r & 0 & 0 & 0 & r & 0 \\ 1-r & 0 & 0 & 0 & 0 & r \\ 1-r & 0 & 0 & 0 & 0 & r \end{bmatrix}$$

